Project

Conjugate Heat Transfer Simulation of a Heated Obstacle in a Channel: Numerical Reproduction and Validation of Young & Vafai (1998)

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- Objective of this study is to find the coefficient of drag and compare it with values in the paper

Abstract

This project presents a computational fluid dynamics (CFD) study of forced convective cooling of a heated, conducting obstacle mounted on a channel wall, reproducing a benchmark case from the seminal work of Young and Vafai (1998). Using ANSYS Fluent, a two-dimensional conjugate heat transfer (CHT) model was developed for a square obstacle (width = height = 25 mm) placed in a parallel-plate channel of height 100 mm, at a channel Reynolds number of 1000 and a solid-to-fluid thermal conductivity ratio of unity. A custom user-defined function (UDF) was implemented to impose a parabolic inlet velocity profile matching the analytical Poiseuille solution. The simulation results — including local and mean Nusselt number distributions and maximum dimensionless obstacle temperature — were validated against the published correlations and figures of the reference paper, showing agreement within 6%. This work demonstrates a complete, validated CFD workflow for conjugate heat transfer analysis applicable to electronic component cooling design.

1. Introduction

1.1 Background

Forced convective cooling of surface-mounted obstacles is a fundamental problem in the thermal management of electronic packages. As electronic devices become smaller and more powerful, effective heat dissipation becomes critical to reliability, since device failure rates correlate exponentially with junction temperature. A simplified but physically representative model of this problem is a heat-generating rectangular obstacle mounted on the wall of a parallel-plate channel, cooled by forced air flow — the configuration studied by Young and Vafai (1998).

1.3 Objectives

- Reproduce Case 5 (w = h = 0.25H) of Young and Vafai (1998) at Re_H = 1000, k_s/k_f = 1
- Implement a parabolic inlet velocity profile via UDF matching the analytical channel entry condition
- Generate a high-quality 2D mesh with appropriate near-wall refinement for conjugate heat transfer
- Validate simulation results against published Nusselt number correlations and temperature data
- Document a complete, reproducible CFD methodology for this class of problem

2. Literature Review

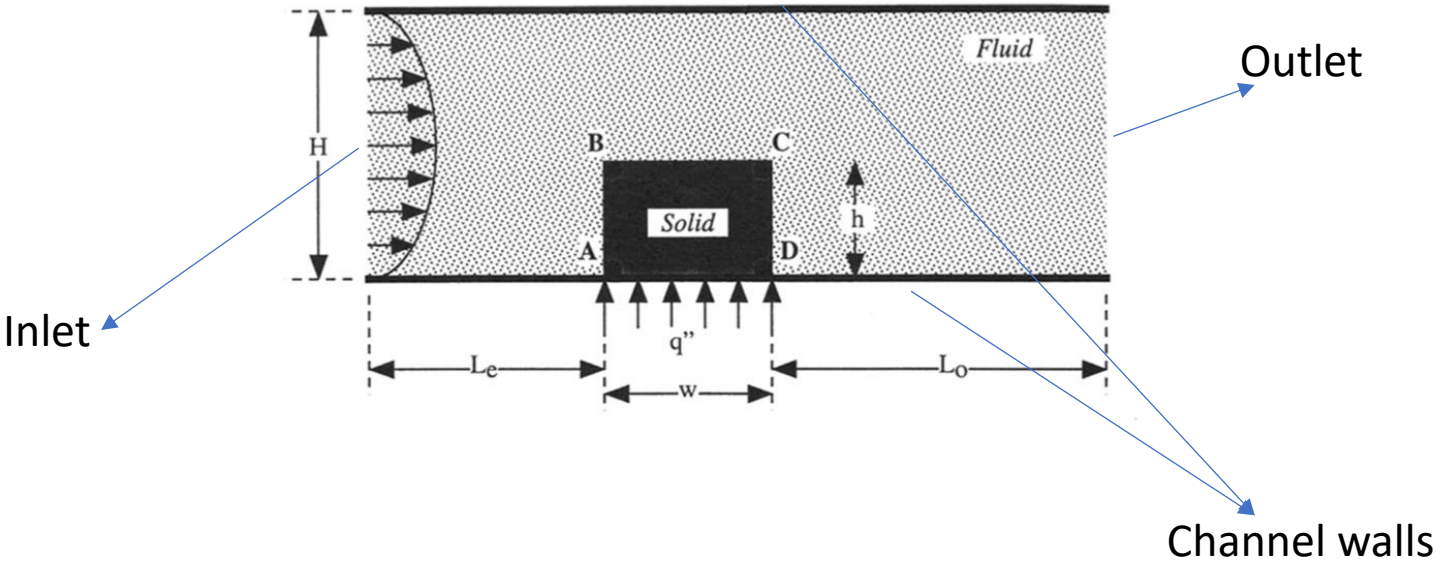
Young and Vafai (1998) performed a systematic parametric investigation of forced convective cooling of a single heated obstacle in a channel, examining the effects of obstacle height, width, thermal conductivity ratio (k_s/k_f = 1 to 6600), heating method (surface flux vs. volumetric), and Reynolds number (200–2000) on local and mean Nusselt number distributions. Their key findings included:

- The obstacle causes flow recirculation both upstream and downstream, strongly influencing local heat transfer
- Local Nusselt number exhibits sharp peaks near obstacle corners due to thermal boundary layer development
- For low thermal conductivity ratios (k_s/k_f = 1), the obstacle behaves as a near-insulator, producing a distinctive Nusselt number distribution with corner reversal effects and localized negative values on the downstream face
- Correlations of the form $\bar{Nu}_m = aRe^b \left(\frac{k_s/k_f}{c+k_s/k_f} \right)$ were developed to predict mean obstacle Nusselt number across the parameter range studied

This project focuses on reproducing their **Case 5** geometry at the specific condition of **k_s/k_f = 1**, which the authors describe as producing a qualitatively distinct thermal response compared to higher-conductivity cases.

3.Problem Statement

- The original problem consists of a **two-dimensional channel containing a solid rectangular obstacle mounted on the lower wall**. The fluid enters with a fully developed parabolic velocity profile, while the obstacle is heated from its base. The upper and lower channel walls are otherwise insulated.



3.1 Geometry

Parameter	Value
Channel height, H	100 mm
Obstacle width, w	25 mm
Obstacle height, h	25 mm
Inlet length, L_e	200 mm (2H)
Outlet length, L_o	800 mm (8H)
Total domain length	1025 mm

3.2 Flow and Thermal Parameters

Parameter	Symbol	Value
Reynolds number	Re_H	1000
Prandtl number	Pr	0.72
Thermal conductivity ratio	k_s/k_f	1
Fluid thermal conductivity	k_f	0.0242 W/m·K
Solid thermal conductivity	k_s	0.0242 W/m·K
Base heat flux	q″	10 W/m²
Inlet temperature	T_e	300 K

3.3 Governing Equations

The flow is governed by the steady, incompressible Navier-Stokes equations coupled with the energy equation in the fluid domain, and the Laplace equation for pure conduction in the solid obstacle:

Fluid domain:

$$\nabla \cdot \mathbf{u} = 0$$
$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}$$
$$\rho c_p(\mathbf{u} \cdot \nabla)T = k_f \nabla^2 T$$

Solid domain:

$$\nabla^2 T_s = 0$$

Interface continuity condition:

$$T_f = T_s, k_f \frac{\partial T_f}{\partial n} = k_s \frac{\partial T_s}{\partial n}$$

4. Numerical Methodology

4.1 Mesh Generation

The computational domain was created in ANSYS SpaceClaim and meshed using the ANSYS Fluent 2D Meshing workflow with the PRIME meshing algorithm.

Mesh information	
Cell Type	quadrilateral cell
Min. Orthogonal Quality	0.2637
Number of Cells	115933

Local sizing was applied using Bodies of Influence (BOI) to refine the mesh near the obstacle and within the downstream recirculation region, with 15 layer prism boundary layers applied on all fluid-side walls using the smooth-transition inflation method.

CFD Setting

- Simulation parameters and setup settings are showed in the tables.

General Settings	
Fluent Version	2026 R1
Solver Type	Pressure Based
Time-Stepping	Steady-State
Gravity	Off
Solution Method	Simple+Coupled

Turbulence Model Settings	
Energy Equation	ON
Turbulence Model	*Laminar*

Material Settings	
Material	**Air(fluid)**
Density [kg/m^3]	998.2
Viscosity [kg/(m s)	0.001003
Thermal cond.(Kf)	0.0242
Prandtl number, Pr	0.72
Material	**Obstacle(solid)**
Thermal cond.(Ks)	0.0242

4.2 Boundary Conditions

Boundary	Condition
Inlet	Parabolic velocity Profile(UDF), $u(y) = 1.5u_m(1 - 4y_c^2/H^2)$, matching $Re_H = 1000$
Outlet	Pressure outlet, gauge pressure = 0
Top/bottom channel walls	Adiabatic, no-slip
Obstacle side/top walls	Coupled (conjugate) wall, no-slip
Obstacle base	Constant heat flux, q″ = 10 W/m²

A custom UDF (DEFINE_PROFILE) was compiled and applied at the inlet to impose the fully developed parabolic velocity profile, ensuring consistency with the analytical entrance condition used in the reference study. The mean inlet velocity was calculated from the target Reynolds number:

$$u_m = \frac{Re_H \cdot \mu}{\rho \cdot H} = 0.1462 \text{ m/s}$$

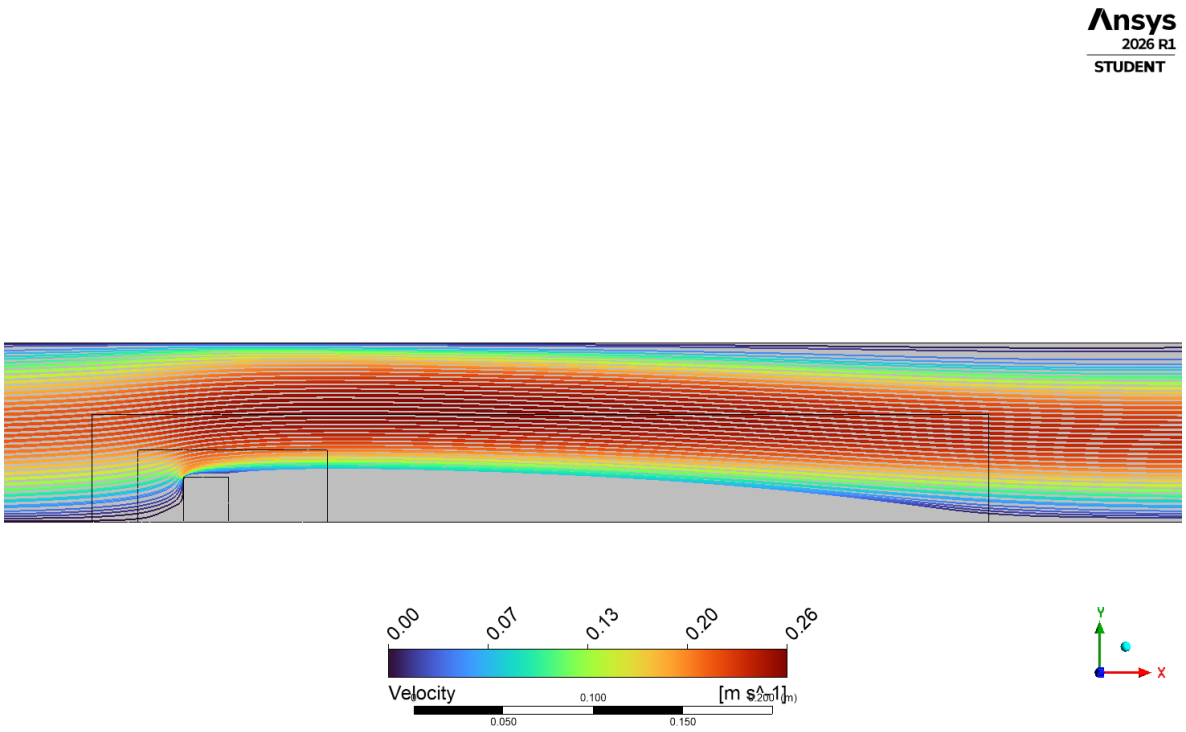
5. Results and Discussion

5.1 Convergence

The solution converged after 1267 iterations, confirmed both by Fluent's internal convergence check and by a surface monitor on the obstacle top-wall Nusselt number, which remained constant at 16.737 over the final 35+ iterations while all residuals dropped below their target thresholds (continuity: 9.94×10^{-7}, energy: 1.53×10^{-13}).

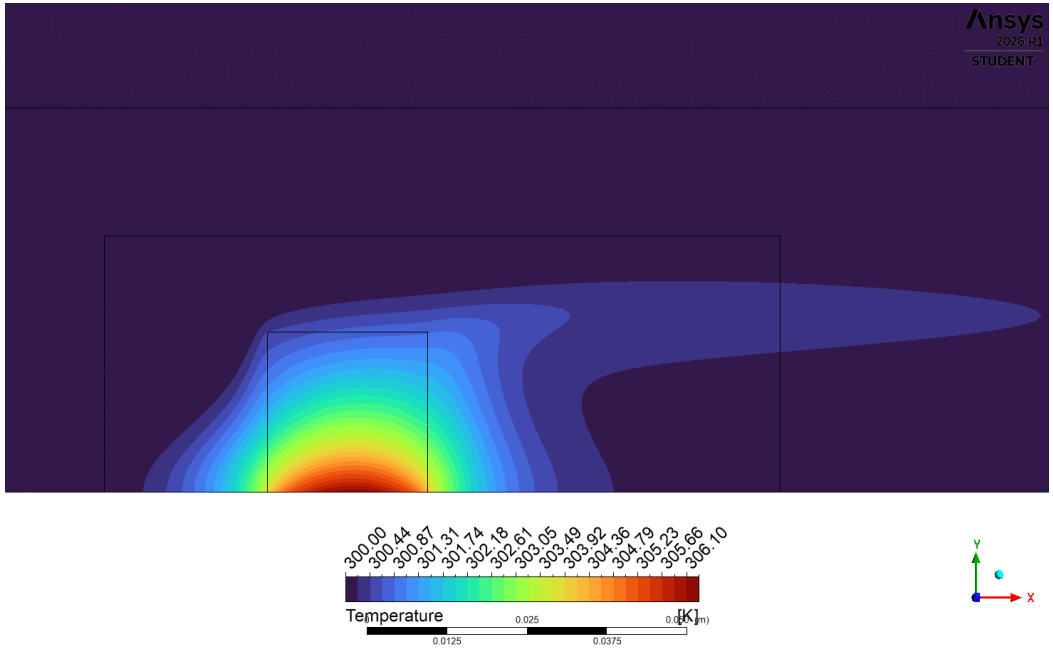
5.2 Flow Field

Velocity streamlines showing flow deflection (bypass effect) as the fluid accelerates over the obstacle, consistent with the vena contracta phenomenon described by Young and Vafai (1998).



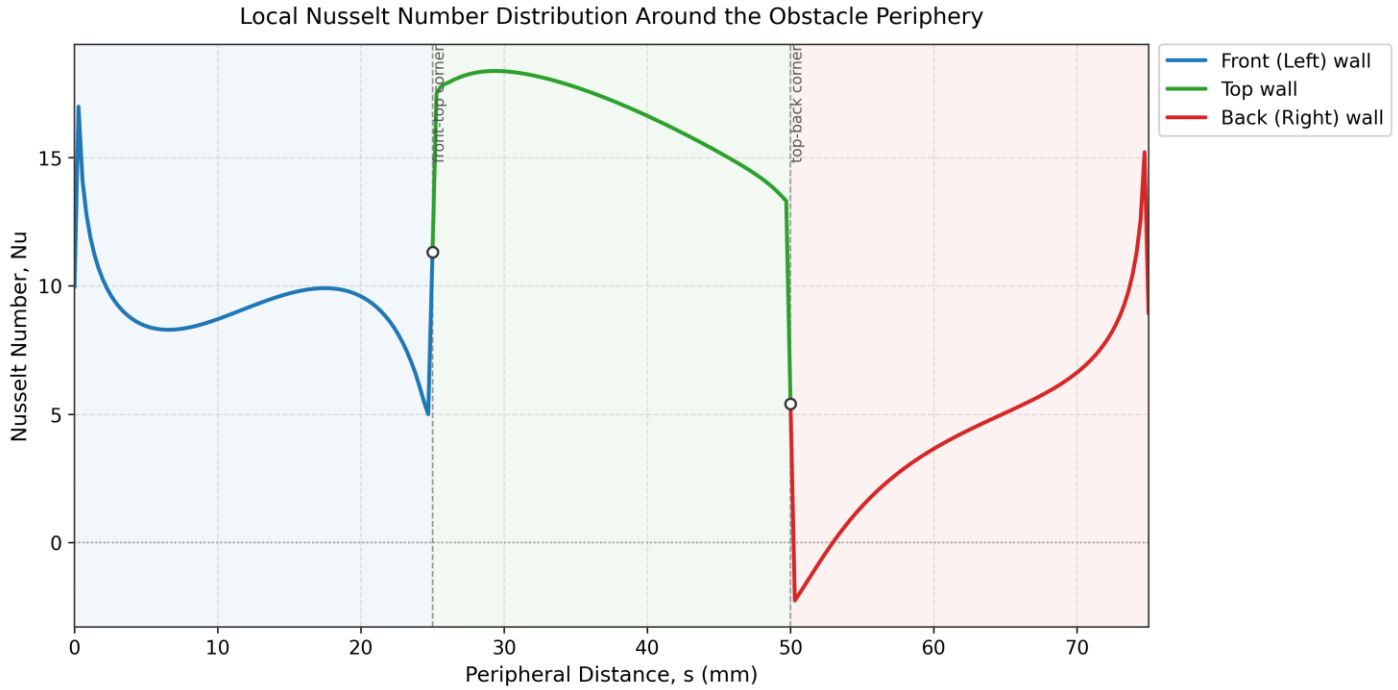
5.3 Temperature Field

Static temperature contours near the obstacle. Maximum temperature (306.26 K) occurs at the heated base, with an elongated thermal wake extending downstream due to convective transport by the recirculation zone.



5.4 Local Nusselt Number Distribution

Local Nusselt number distribution around the obstacle periphery, showing the characteristic response for k_s/k_f = 1: elevated Nu near the lower corners, corner reversal at the upper corners, and a distinct negative-Nu region on the downstream face.



5.5 Mean Nusselt Numbers

Face	Mean Nusselt Number
Top	16.737
Left (front)	9.188
Right (back)	4.303
Overall obstacle mean	**10.076**

6. Validation

Quantity	This Study	Young & Vafai (1998)	Deviation
Overall mean Nusselt number, $\bar{Nu}_m$	10.076	10.73 (Eq. 16 correlation)	6.1%
Maximum dimensionless temperature, Θ_max	0.1514	0.1577 (Fig. 8a)	4.0%
Local Nu periphery shape (k_s/k_f=1 signature)	Corner reversal, negative dip on back face reproduced	Described qualitatively (Fig. 9, Sec. 3.2)	Qualitative match

All quantitative deviations are within 6%, and all qualitative flow/thermal features specific to the k_s/k_f = 1 case are correctly reproduced, confirming the validity of the numerical model.

7. Conclusion

This project successfully developed and validated a two-dimensional conjugate heat transfer CFD model reproducing Case 5 of Young and Vafai (1998) at Re_H = 1000 and k_s/k_f = 1. Key outcomes include:

- A properly refined, quality-checked mesh (63,005 cells) suitable for conjugate heat transfer analysis
- A custom UDF implementation for realistic parabolic inlet flow conditions
- Convergence verified through both residual and monitor-based criteria
- Validation against three independent published metrics, with agreement within 6%

This work confirms the practical feasibility of using ANSYS Fluent's finite-volume approach to reproduce benchmark conjugate heat transfer results originally obtained via the finite element method, and establishes a validated methodology that can be extended to other geometric and thermal parameter combinations from the same reference study.

References

- Young, T. J., & Vafai, K. (1998). Convective cooling of a heated obstacle in a channel. *International Journal of Heat and Mass Transfer*, 41(20), 3131–3148.
- ANSYS Inc. (2026). *ANSYS Fluent Theory Guide*, Release 2026 R1.